Introduction to Analysis

FMA_1F002_EP

Lecture Notes

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These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

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1 Number Systems

1.1 Natural Numbers, Integers, and Rationals

Definition 1: Natural Numbers

The set of natural numbers $\mathbb{N}$ is characterized by the following axioms:

1. $0 \in \mathbb{N}$.
2. If $n \in \mathbb{N}$, then $n + 1 \in \mathbb{N}$.

The Principle of Mathematical Induction states that if a subset $E \subset \mathbb{N}$ satisfies these two properties, then $E = \mathbb{N}$.

Definition 2: Bounded Subsets of $\mathbb{N}$

Let $E \subset \mathbb{N}$. $m \in E$ is the least element of E if $m \leq n$ for all $n \in E$. A subset $E \subset \mathbb{N}$ is said to be bounded if there exists M such that for all $m \in E$, $m \leq M$. $M \in \mathbb{N}$ is said to be the largest (longest) element of $E \subset \mathbb{N}$ if $M \in E$ and for all $n \in E, n \leq M$.

Proposition 3: Existence of Extrema in $\mathbb{N}$ and $\mathbb{Z}$

- Let $E \subset \mathbb{N}$ be non-empty. Then E has a unique least element.
- A bounded non-empty subset $E \subset \mathbb{N}$ has a unique largest element.
- Let $E \subset \mathbb{Z}$ be non-empty and bounded from above (respectively, below). Then E has a largest (respectively, least) element.

Proof. (Uniqueness of the least element): Suppose m_1 and m_2 are both least elements of $E \subset \mathbb{N}$. Since m_1 is a least element and $m_2 \in E$, $m_1 \leq m_2$. Symmetrically, since m_2 is a least element and $m_1 \in E$, $m_2 \leq m_1$. Thus, $m_1 = m_2$. The existence is guaranteed by the Well-Ordering Principle, which is equivalent to the Axiom of Induction. □

Remark 4: Algebraic Structures

The integers $(\mathbb{Z}, +, \times, \leq)$ form an ordered commutative ring, where $\leq$ is a total order compatible with $+$ and $\times$. The rational numbers $(\mathbb{Q}, +, \times, \leq)$ form an ordered commutative field. **Important Warning:** $\mathbb{Q}$ does not satisfy the least upper bound property (e.g., the set $\{q \in \mathbb{Q} : q^2 < 2\}$ has no supremum in $\mathbb{Q}$).

Theorem 5: Properties of the Rationals $\mathbb{Q}$

- **Archimedean Property:** If $q \in \mathbb{Q}$ and $q > 0$, then there exists $m, n \in \mathbb{N}$ such that $\frac{1}{m} \leq q \leq n$.
- **Irrationality of $\sqrt{2}$:** For all $q \in \mathbb{Q}$, $q^2 \neq 2$.

(Note: Two integers a and b are said to be coprime if their greatest common divisor is 1, denoted $a \wedge b = 1 \iff (a, b) = 1$).

PROOF. (Irrationality of $\sqrt{2}$): Suppose for contradiction that there exists $q \in \mathbb{Q}$ such that $q^2 = 2$. We can write $q = \frac{a}{b}$ where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a, b) = 1$ (the fraction is in simplest form). Then:

$$\left(\frac{a}{b}\right)^2 = 2 \implies a^2 = 2b^2$$

This implies a^2 is even. Since the square of an odd number is odd, a must also be even. We can write $a = 2k$ for some integer k. Substituting this into the equation gives:

$$(2k)^2 = 2b^2 \implies 4k^2 = 2b^2 \implies 2k^2 = b^2$$

This implies b^2 is even, which means b is also even. But if both a and b are even, they share a common factor of 2, which contradicts our initial assumption that $\gcd(a, b) = 1$. Thus, no such q exists. □

1.2 Real Numbers and Exponentiation

Definition 6: Bounds and Supremum/Infimum

Let $E \subset \mathbb{R}$. We say that M is an upper bound of E if $x \leq M$ for all $x \in E$. We say M is the least upper bound (denoted $\sup E$) if M is an upper bound of E and any upper bound M' of E satisfies $M \leq M'$. The infimum ($\inf E$) is defined similarly as the greatest lower bound.

Proposition 7: Sequential Characterization of Supremum

For $E \subset \mathbb{R}$, $M = \sup E$ if and only if M is an upper bound of E and there exists a sequence $(a_n)_{n \geq 0} \subset E$ such that $\lim_{n \to \infty} a_n = M$.

Theorem 8: Properties of Reals $\mathbb{R}$

$(\mathbb{R}, +, \times, \leq)$ is an ordered commutative field that enjoys the "least upper bound property". Let $E \subset \mathbb{R}$ be a nonempty subset. $M = \sup E$ if and only if M is an upper bound and for all $\epsilon > 0$, there exists $x \in E$ such that $M - \epsilon < x \leq M$.

- **Archimedean Property:** For all $x \in \mathbb{R}$ with $x > 0$, there exists $m, n \in \mathbb{N}$ such that $\frac{1}{m} < x < n$.
- **Integer Part:** Given $x \in \mathbb{R}$, there exists a unique integer k such that $k \leq x < k + 1$.

PROOF. (Archimedean Property): Suppose for contradiction that there exists $x > 0$ such that for all $n \in \mathbb{N}$, $nx \leq y$ for some fixed $y \in \mathbb{R}$. This means the set $A = \{nx : n \in \mathbb{N}\}$ is bounded above by y. By the least upper bound property of $\mathbb{R}$, A has a supremum, say $M = \sup A$. Since $x > 0$, $M - x < M$. Therefore, $M - x$ is not an upper bound for A. There must exist some element $mx \in A$ (where $m \in \mathbb{N}$) such that $mx > M - x$. But then:

$$mx + x > M \implies (m + 1)x > M$$

Since $m + 1 \in \mathbb{N}$, $(m + 1)x \in A$, which contradicts that M is an upper bound for A. Thus, $\mathbb{R}$ is Archimedean. □

Theorem 9: Existence of Roots

- $\sqrt{2}$ exists in $\mathbb{R}$.
- Given $n \geq 2$ and $x \in \mathbb{R}$ with $x > 0$, there exists a unique $\alpha \in \mathbb{R}$ with $\alpha \geq 0$ such that $\alpha^n = x$. We denote this as $\alpha = x^{1/n}$ or $\alpha = \sqrt[n]{x}$.

PROOF. (Existence of $\sqrt{2}$): Let $E = \{x \in \mathbb{R} : x^2 \leq 2, x \geq 0\}$. E is non-empty ($1 \in E$) and bounded above (by 2). By the least upper bound property, $\alpha = \sup E$ exists. If $\alpha^2 < 2$: For small $\epsilon > 0$, $(\alpha + \epsilon)^2 = \alpha^2 + 2\alpha\epsilon + \epsilon^2$. We can choose ϵ small enough such that this is still < 2, meaning $\alpha + \epsilon \in E$, contradicting α being an upper bound. If $\alpha^2 > 2$: For small $\epsilon > 0$, $(\alpha - \epsilon)^2 = \alpha^2 - 2\alpha\epsilon + \epsilon^2$. We can choose ϵ such that this is > 2. Then $\alpha - \epsilon$ is an upper bound strictly less than α, contradicting it being the *least* upper bound. Hence, $\alpha^2 = 2$.
(Existence of n-th root): Let $x > 0$ and $E = \{y \in \mathbb{R} : y \geq 0, y^n \leq x\}$. E is bounded above (by $\max(1, x)$) and non-empty ($0 \in E$). Let $\alpha = \sup E$. By a similar argument using the binomial expansion of $(\alpha \pm \epsilon)^n$, assuming $\alpha^n < x$ allows us to find $\alpha + \epsilon \in E$, and $\alpha^n > x$ allows us to find an upper bound $\alpha - \epsilon$, both leading to contradictions. Thus, $\alpha^n = x$. Uniqueness follows because the function $f(y) = y^n$ is strictly increasing on $[0, +\infty)$. □

Definition 10: Exponentiation

For $x > 0$, we define $x^0 = 1$ and by induction $x^{n+1} = x \cdot x^n$ and $x^{-n} = \frac{1}{x^n}$ for $n \in \mathbb{N}$. For a rational $r = p/q \in \mathbb{Q}$ ($p \in \mathbb{Z}, q \in \mathbb{N}_{>0}$), we define $x^r = x^{p/q} = \sqrt[q]{x^p} = (\sqrt[q]{x})^p$. Main properties for $x, y > 0$ and $r, r' \in \mathbb{Q}$:

- $x^r \cdot x^{r'} = x^{r+r'}$ and $(x^r)^{r'} = x^{r \cdot r'}$.
- If $0 < x < y$, then $x^r < y^r$ if $r > 0$, and $x^r > y^r$ if $r < 0$.
- If $r < r'$, then $x^r < x^{r'}$ if $x > 1$, and $x^r > x^{r'}$ if $x < 1$.

1.3 Density, Absolute Value, and Intervals

Definition 11: Density

A subset $E \subset \mathbb{R}$ is said to be dense in $\mathbb{R}$ if for all $x, y \in \mathbb{R}$ with $x < y$, there exists $e \in E$ such that $x < e < y$. Equivalently, E is dense if for all $x \in \mathbb{R}$ and all $\epsilon > 0$, there exists $e \in E$ such that $|e - x| < \epsilon$.

Theorem 12: Density of $\mathbb{Q}$

$\mathbb{Q}$ is dense in $\mathbb{R}$. (The set of irrational numbers is also dense in $\mathbb{R}$).

PROOF. Let $x, y \in \mathbb{R}$ with $x < y$. Since $y - x > 0$, by the Archimedean property, there exists an integer $n \in \mathbb{N}$ such that $n(y - x) > 1$, which means $ny - nx > 1$. Thus, the interval (nx, ny) has a length strictly greater than 1. It must contain at least one integer. Let $m \in \mathbb{Z}$ be the integer such that $m - 1 \leq nx < m$. Since $nx < m$, and the interval length is > 1, we also have $m < ny$. Dividing the inequalities by n (since $n > 0$) yields:

$$x < \frac{m}{n} < y$$

Since $\frac{m}{n} \in \mathbb{Q}$, we have found a rational number strictly between x and y. □

Problem 13: Density of Irrationals

Prove that the set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

Definition 14: Absolute Value

For $x \in \mathbb{R}$, $|x| = x$ if $x \geq 0$, and $-x$ if $x < 0$. Properties:

1. $|x| = 0$ if and only if $x = 0$.
2. $-|x| \leq x \leq |x|$.
3. **Triangle inequality:** $|x + y| \leq |x| + |y|$.
4. $||y| - |x|| \leq |y - x|$.
5. If $-x \leq y \leq x$, then $|y| \leq x$.

PROOF. (Triangle Inequality): For any $x, y \in \mathbb{R}$, we have:

$$-|x| \leq x \leq |x| \quad \text{and} \quad -|y| \leq y \leq |y|$$

Adding these two inequalities together yields:

$$-(|x| + |y|) \leq x + y \leq |x| + |y|$$

This is equivalent to $|x + y| \leq |x| + |y|$. □

Definition 15: Intervals

A subset $X \subset \mathbb{R}$ is an interval if for all $x, y \in X$ with $x \leq y$, then $[x, y] \subset X$. Intervals take the forms: $\emptyset, \mathbb{R}, [a, b], (a, b], (a, b), [a, b), (-\infty, a], (-\infty, a), (a, +\infty), [a, +\infty)$.

Remark 16: Basic Trigonometric Inequalities

For $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $|\sin x| \leq |x| \leq |\tan x|$.

1.4 Complex Numbers

Definition 17: Complex Plane

The set of complex numbers is $\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}$ where $i^2 = -1$. $(\mathbb{C}, +, \times)$ is a commutative field. For $z = x + iy \in \mathbb{C}$:

- **Complex conjugate:** $\overline{z} = x - iy$.
- $z \cdot \overline{z} = x^2 + y^2$.
- **Real and Imaginary parts:** $x = \text{Re}(z) = \frac{z+\overline{z}}{2}$, $y = \text{Im}(z) = \frac{z-\overline{z}}{2i}$.
- **Modulus:** $|z| = \sqrt{z\overline{z}}$. Properties: $|z| = 0 \iff z = 0$, $|z| = |\overline{z}|$, $|z_1 z_2| = |z_1||z_2|$, and $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle Inequality).

Theorem 18: Euler's and de Moivre's Formulas

Any non-zero complex number can be written as $z = |z|(\cos\theta + i\sin\theta)$ where $\theta \in (-\pi, \pi]$ is the principal argument.

- **Euler's Formula:** $e^{i\theta} = \cos\theta + i\sin\theta$. Thus, $e^{i\theta} \cdot e^{i\varphi} = e^{i(\varphi+\theta)}$.
- **de Moivre's formula:** Let $z = \rho e^{i\theta}$ $(\rho \geq 0, \theta \in \mathbb{R})$. For $n \in \mathbb{Z}$, $z^n = \rho^n e^{in\theta}$.

PROOF. (de Moivre's Formula for $n \in \mathbb{N}$): Proceed by induction. For $n = 1$, it is trivially true. Assume it holds for $n = k$: $(\cos\theta + i\sin\theta)^k = \cos(k\theta) + i\sin(k\theta)$. For $n = k + 1$:

$$\begin{aligned}(\cos\theta + i\sin\theta)^{k+1} &= (\cos(k\theta) + i\sin(k\theta))(\cos\theta + i\sin\theta) \\ &= (\cos(k\theta)\cos\theta - \sin(k\theta)\sin\theta) + i(\sin(k\theta)\cos\theta + \cos(k\theta)\sin\theta) \\ &= \cos((k+1)\theta) + i\sin((k+1)\theta)\end{aligned}$$

The property extends to negative integers via inverse multiplication. □

Theorem 19: n-th Roots of Unity

For $n \geq 1$, the solutions to $z^n = 1$ in $\mathbb{C}$ are given by $1, \omega, \omega^2, \ldots, \omega^{n-1}$ where $\omega = e^{i\frac{2\pi}{n}}$. More generally, the solutions to $z^n = u$ (where $u = \rho e^{i\theta}$) are $z_k = \sqrt[n]{\rho} e^{i\frac{\theta}{n}} e^{i\frac{2\pi}{n}k}$ for $k = 0, 1, \ldots, n-1$.

PROOF. To solve $z^n = 1$, we express z in polar form: $z = re^{i\theta}$ where $r \geq 0$. Then $z^n = r^n e^{in\theta}$. We also express $1 = 1e^{i(0+2\pi k)}$ for $k \in \mathbb{Z}$. Equating the moduli and arguments: 1) $r^n = 1 \implies r = 1$ (since r is a non-negative real number). 2) $n\theta = 2\pi k \implies \theta = \frac{2\pi k}{n}$. The distinct values of $e^{i\frac{2\pi k}{n}}$ occur for $k = 0, 1, 2, \ldots, n-1$. Letting $\omega = e^{i\frac{2\pi}{n}}$, the roots are $1, \omega, \omega^2, \ldots, \omega^{n-1}$. For $z^n = u = \rho e^{i\theta}$, we have $r^n e^{in\phi} = \rho e^{i(\theta+2\pi k)}$. Thus $r = \sqrt[n]{\rho}$ and $\phi = \frac{\theta}{n} + \frac{2\pi k}{n}$ for $k = 0, \ldots, n-1$. □

Remark 20: Principal Square Root

The principal square root of $z = \rho e^{i\theta}$ $(\theta \in (-\pi, \pi])$ is $\sqrt{z} = \sqrt{\rho} e^{i\frac{\theta}{2}}$. **Mystery / Fallacy of Complex Roots:** Consider $\sqrt{-1} = i$ and $\sqrt{1} = 1$. Notice that $\sqrt{(-1)(-1)} = \sqrt{1} = 1$. However, if we split the root, $\sqrt{-1}\sqrt{-1} = i \cdot i = -1$. Thus $1 \neq -1$. This step-by-step counterexample shows that the property $\sqrt{ab} = \sqrt{a}\sqrt{b}$ fails for complex numbers.

2 Sequences

Definition 21: Sequences

A sequence with values in $\mathbb{K}$ ($\mathbb{R}$ or $\mathbb{C}$) and indexes in $I \subset \mathbb{N}$ is a function $I \to \mathbb{K}$, denoted by $(a_n)_{n\in I}$. Sequences can be described explicitly (e.g., $u_n = \frac{n+(-1)^n}{n-(-1)^n}$) or inductively (e.g., $u_0 = 1, u_{n+1} = \frac{1}{2}(u_n + \frac{2}{u_n})$). A sequence is bounded if there exists $M > 0$ such that $|a_n| \leq M$ for all n. Real sequences can be bounded from above ($a_n \leq M$) or below ($a_n \geq M$).

Remark 22: Sequence vs. Set of Values Warning

A sequence $(a_n)_{n\geq 0}$ must not be confused with its range/set of values $\{a_n : n \in \mathbb{N}\}$. For example, if $a_n = (-1)^n$, the sequence is $(1, -1, 1, -1, \dots)$, but its set of values is simply $\{-1, 1\}$.

Definition 23: Limits of Sequences

A $\mathbb{K}$-valued sequence $(a_n)_{n\geq 0}$ converges to a limit $l \in \mathbb{K}$ as $n \to +\infty$ if:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, |a_n - l| < \epsilon$$

We write $l = \lim_{n\to\infty} a_n$. Otherwise, the sequence diverges. **Negation of limit definition:** The formal logical negation of sequence convergence to l is:

$$\exists \epsilon > 0, \forall N \in \mathbb{N}, \exists n \geq N : |a_n - l| \geq \epsilon$$

Example 1: Limit Example

Prove $a_n = \frac{(-1)^n}{n+1} \to 0$. Given $\epsilon > 0$, we want:

$$\left|\frac{(-1)^n}{n+1} - 0\right| < \epsilon \iff \frac{1}{n+1} < \epsilon \iff n+1 > \frac{1}{\epsilon} \iff n > \frac{1}{\epsilon} - 1$$

By the Archimedean property, we can choose an integer $N > \frac{1}{\epsilon} - 1$. Then for all $n \geq N$, $|a_n - 0| < \epsilon$.

Lemma 24: Properties of Convergent Sequences

If a sequence converges, its limit is unique. If a sequence converges, it is bounded.

PROOF. (Uniqueness): Suppose (a_n) converges to two limits, l_1 and l_2, with $l_1 \neq l_2$. Let $\epsilon = \frac{|l_1 - l_2|}{2} > 0$. There exists N_1 such that $\forall n \geq N_1, |a_n - l_1| < \epsilon$. There exists N_2 such that $\forall n \geq N_2, |a_n - l_2| < \epsilon$. For $n \geq \max(N_1, N_2)$, by the triangle inequality:

$$|l_1 - l_2| = |(l_1 - a_n) + (a_n - l_2)| \leq |a_n - l_1| + |a_n - l_2| < \epsilon + \epsilon = |l_1 - l_2|$$

This yields $|l_1 - l_2| < |l_1 - l_2|$, a contradiction. Thus $l_1 = l_2$.
(Boundedness): Let $\lim a_n = l$. Choosing $\epsilon = 1$, there exists $N \in \mathbb{N}$ such that $\forall n \geq N$, $|a_n - l| < 1$. By the triangle inequality, $|a_n| = |(a_n - l) + l| \leq |a_n - l| + |l| < 1 + |l|$ for all $n \geq N$. Let $M = \max(|a_0|, |a_1|, \dots, |a_{N-1}|, 1 + |l|)$. Then for all $n \in \mathbb{N}$, $|a_n| \leq M$. Hence, the sequence is bounded. □

Theorem 25: Algebra of Limits

Assume (a_n) and (b_n) converge to a and b respectively. Then $(a_n + b_n)$ converges to $a + b$, and $(a_n b_n)$ converges to ab. If $b \neq 0$ and $b_n \neq 0$, then (a_n/b_n) converges to a/b. Furthermore, a complex sequence (a_n) converges if and only if (Re a_n) and (Im a_n) both converge.

PROOF. (Sum): Let $\epsilon > 0$. There exists N_1 such that $\forall n \geq N_1, |a_n - a| < \epsilon/2$. There exists N_2 such that $\forall n \geq N_2, |b_n - b| < \epsilon/2$. For $n \geq \max(N_1, N_2)$:

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus $\lim(a_n + b_n) = a + b$.
(Product): We write $a_n b_n - ab = a_n b_n - a_n b + a_n b - ab = a_n(b_n - b) + b(a_n - a)$. Since (a_n) converges, it is bounded by some $M > 0$. Let $\epsilon > 0$. Choose N large enough such that for $n \geq N$, $|a_n - a| < \frac{\epsilon}{2(|b|+1)}$ and $|b_n - b| < \frac{\epsilon}{2M}$. Then:

$$|a_n b_n - ab| \leq |a_n||b_n - b| + |b||a_n - a| < M\frac{\epsilon}{2M} + |b|\frac{\epsilon}{2(|b|+1)} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus $\lim(a_n b_n) = ab$. □

Theorem 26: Squeeze Theorem and Inequalities

If (a_n) and (b_n) are real-valued convergent sequences with $a_n \leq b_n$, then $\lim a_n \leq \lim b_n$.
Squeeze theorem: Let $(a_n), (b_n), (c_n)$ be real-valued such that $a_n \leq b_n \leq c_n$. If (a_n) and (c_n) converge to l, then (b_n) converges to l.

PROOF. (Squeeze Theorem): Let $\epsilon > 0$. Since $\lim a_n = l$, there exists N_1 such that $\forall n \geq N_1$, $l - \epsilon < a_n$. Since $\lim c_n = l$, there exists N_2 such that $\forall n \geq N_2$, $c_n < l + \epsilon$. Let $N = \max(N_1, N_2)$. For all $n \geq N$:

$$l - \epsilon < a_n \leq b_n \leq c_n < l + \epsilon$$

This implies $l - \epsilon < b_n < l + \epsilon$, meaning $|b_n - l| < \epsilon$. Thus, $\lim b_n = l$. □

Definition 27: Infinite Limits

A sequence (a_n) tends to $+\infty$ if $\forall A > 0, \exists N \in \mathbb{N}$ such that $\forall n \geq N$, $a_n > A$. If $a_n \leq b_n$ and $\lim a_n = +\infty$, then $\lim b_n = +\infty$.

Theorem 28: Monotone Sequences

A real-valued sequence (a_n) is increasing if $a_{n+1} \geq a_n$, strictly increasing if $a_{n+1} > a_n$, and decreasing if $a_{n+1} \leq a_n$. If (a_n) is increasing and bounded from above, then (a_n) converges.

PROOF. Let $E = \{a_n : n \in \mathbb{N}\}$. Since the sequence is bounded from above, E has a supremum $l = \sup E$. Let $\epsilon > 0$. Since l is the least upper bound, $l - \epsilon$ is not an upper bound for E. There exists some $N \in \mathbb{N}$ such that $a_N > l - \epsilon$. Because (a_n) is increasing, for all $n \geq N$, we have $a_n \geq a_N > l - \epsilon$. Because l is an upper bound, $a_n \leq l < l + \epsilon$ for all n. Combining these, $l - \epsilon < a_n < l + \epsilon$ for all $n \geq N$, which means $|a_n - l| < \epsilon$. Hence, $\lim a_n = l$. □

Theorem 29: Adjacent Sequences

Two sequences (a_n) and (b_n) are adjacent if (a_n) is increasing, (b_n) is decreasing, and $\lim(b_n - a_n) = 0$. If (a_n) and (b_n) are adjacent, both converge to the same limit.

PROOF. Consider the sequence $d_n = b_n - a_n$. $d_{n+1} - d_n = (b_{n+1} - b_n) - (a_{n+1} - a_n)$. Since $b_{n+1} \leq b_n$ and $a_{n+1} \geq a_n$, both parenthetical terms are non-positive, so $d_{n+1} \leq d_n$. Since

$d_n \to 0$ and is decreasing, we must have $d_n \geq 0$ for all n, meaning $a_n \leq b_n$. Therefore, (a_n) is increasing and bounded above by b_0 (since $a_n \leq b_n \leq b_0$), so it converges to some limit l_1. Similarly, (b_n) is decreasing and bounded below by a_0, so it converges to some limit l_2. By the algebra of limits, $\lim(b_n - a_n) = l_2 - l_1$. But we are given this limit is 0. Thus $l_1 = l_2$. □

Definition 30: Limit Superior and Inferior

Let (a_n) be a bounded real sequence. Define $b_n = \inf\{a_k : k \geq n\}$ and $c_n = \sup\{a_k : k \geq n\}$. Then $b_n \leq a_n \leq c_n$. The limit superior is $\limsup a_n = \lim_{n\to\infty} c_n$. The limit inferior is $\liminf a_n = \lim_{n\to\infty} b_n$. (a_n) converges if and only if $\limsup a_n = \liminf a_n$.

Definition 31: Subsequences and Accumulation Points

Given a strictly increasing function $\phi : \mathbb{N} \to \mathbb{N}$ (thus $\phi(n) \geq n$), $(a_{\phi(n)})_{n\geq 0}$ is a subsequence of $(a_n)_{n\geq 0}$. $l \in \mathbb{K}$ is an accumulation point of (a_n) if there exists a subsequence converging to l. Equivalently, l is an accumulation point if $\forall \epsilon > 0, \forall N \in \mathbb{N}, \exists n \geq N$ such that $|a_n - l| < \epsilon$.

Remark 32: Properties of Accumulation Points

For $a_n = (-1)^n$, the set of values is $\{-1, 1\}$. The accumulation points are -1 and 1. A convergent sequence has its limit as its unique accumulation point.

Theorem 33: Bolzano-Weierstrass Theorem

Let $(a_n)_{n\geq 0}$ be a bounded real or complex-valued sequence. Then $(a_n)_{n\geq 0}$ has at least one accumulation point. For a real sequence, $\limsup a_n$ is the greatest accumulation point, and $\liminf a_n$ is the lowest accumulation point.

Proof. We prove this for $\mathbb{R}$ using the property that every sequence has a monotonic subsequence. We call an index n a "peak" if $a_n > a_m$ for all $m > n$. Case 1: There are infinitely many peaks. Let $n_1 < n_2 < n_3 < \dots$ be the indices of the peaks. By definition, $a_{n_1} > a_{n_2} > a_{n_3} > \dots$, so we have constructed a strictly decreasing subsequence. Case 2: There are finitely many peaks. Let N be the index of the last peak (or $N = 0$ if there are none). Choose $n_1 > N$. Since n_1 is not a peak, there exists $n_2 > n_1$ such that $a_{n_2} \geq a_{n_1}$. Since n_2 is not a peak, there exists $n_3 > n_2$ such that $a_{n_3} \geq a_{n_2}$. Continuing this process yields an increasing subsequence. In either case, (a_n) has a monotonic subsequence. Since the original sequence is bounded, the monotonic subsequence is also bounded. By the Monotone Convergence Theorem, this subsequence converges to some limit l. Thus, l is an accumulation point of (a_n). □

3 Infinite Series

Definition 34: Infinite Series

For a sequence $a_k \in \mathbb{K}$, the partial sum of order n is defined as $S_n = \sum_{k=0}^{n} a_k$. The infinite series $\sum_{n=0}^{\infty} a_n$ is said to converge if the sequence of partial sums $(S_n)_{n\geq 0}$ converges to a limit as $n \to \infty$. Its sum is denoted $\sum_{k=0}^{\infty} a_k = \lim_{n\to\infty} S_n$. If $\sum a_n$ and $\sum b_n$ converge, then $\sum(a_n + b_n) = \sum a_n + \sum b_n$.

Remark 35: Index Manipulation Rules

For finite sums:

- Index reversal: $\sum_{i=m}^{n} a_i = \sum_{i=m}^{n} a_{n+m-i}$
- Index shifting: $\sum_{j=m}^{n} a_j = \sum_{j=m+p}^{n+p} a_{j-p}$
- Index permutations: $\sum_{k=m}^{n} a_k = \sum_{k=m}^{n} a_{\sigma(k)}$ for any bijection σ.

Lemma 36: Cauchy-Schwarz Inequality

For all real numbers $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$:

$$\left(\sum_{k=1}^{n} x_k y_k\right)^2 \leq \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{j=1}^{n} y_j^2\right)$$

Geometric vector interpretation: $|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\|\|\vec{y}\|$. Equality holds if and only if there exists a real number μ such that $x_k = \mu y_k$ for all $k = 1, \ldots, n$ (or $y_k = 0$).

Proof. Consider the polynomial in a real variable t:

$$P(t) = \sum_{k=1}^{n} (x_k + t y_k)^2$$

Since it is a sum of squares, $P(t) \geq 0$ for all $t \in \mathbb{R}$. Expanding the expression:

$$P(t) = \sum_{k=1}^{n} (x_k^2 + 2t x_k y_k + t^2 y_k^2) = \left(\sum_{k=1}^{n} y_k^2\right) t^2 + 2\left(\sum_{k=1}^{n} x_k y_k\right) t + \left(\sum_{k=1}^{n} x_k^2\right)$$

This is a quadratic polynomial $At^2 + Bt + C \geq 0$. Because it is non-negative everywhere, its discriminant must be less than or equal to zero: $\Delta = B^2 - 4AC \leq 0$.

$$4\left(\sum_{k=1}^{n} x_k y_k\right)^2 - 4\left(\sum_{k=1}^{n} y_k^2\right)\left(\sum_{k=1}^{n} x_k^2\right) \leq 0$$

Dividing by 4 and rearranging yields the Cauchy-Schwarz inequality. Equality holds if and only if $\Delta = 0$, which means $P(t) = 0$ has a root, so $x_k + t y_k = 0$ for all k. □

Proposition 37: Non-Negative Series Convergence Criterion

If $u_n \geq 0$ for all n, then the series $\sum u_n$ converges if and only if its sequence of partial sums $(S_n)_{n \geq 0}$ is bounded from above.

Theorem 38: Convergence Tests

- **Comparison Test:** If $0 \leq u_n \leq v_n$ and $\sum v_n$ converges, then $\sum u_n$ converges. If $\sum u_n$ diverges, then $\sum v_n$ diverges.
- **Geometric Series:** For $a \in \mathbb{C}$, the series $\sum_{n=0}^{\infty} a^n$ converges if and only if $|a| < 1$.
- **Zero Test:** If $\sum a_n$ converges, then $\lim_{n \to \infty} a_n = 0$.

- **Telescoping Series:** $\sum_{n=0}^{\infty}(a_{n+1}-a_n)$ converges if and only if (a_n) converges, and its sum is $\lim_{n\to\infty} a_n - a_0$.

PROOF. (Comparison Test): Let $S_n = \sum_{k=0}^{n} u_k$ and $T_n = \sum_{k=0}^{n} v_k$. Since $u_n \geq 0$ and $v_n \geq 0$, both (S_n) and (T_n) are increasing sequences. If $\sum v_n$ converges to L, then $T_n \leq L$ for all n. Thus $S_n \leq T_n \leq L$. Since (S_n) is increasing and bounded above by L, it converges.
(Geometric Series): Let $S_n = \sum_{k=0}^{n} a^k$. For $a \neq 1$, $S_n = \frac{1-a^{n+1}}{1-a}$. If $|a| < 1$, $a^{n+1} \to 0$, so $S_n \to \frac{1}{1-a}$. If $|a| \geq 1$, the terms do not tend to zero, so it diverges by the Zero Test.
(Zero Test): Let $S_n = \sum_{k=0}^{n} a_k$. Since the series converges, $S_n \to L$. Notice that $a_n = S_n - S_{n-1}$. Taking the limit as $n \to \infty$, $\lim a_n = \lim S_n - \lim S_{n-1} = L - L = 0$.
(Telescoping): Let $S_n = \sum_{k=0}^{n}(a_{k+1}-a_k) = (a_1-a_0)+(a_2-a_1)+\cdots+(a_{n+1}-a_n) = a_{n+1}-a_0$. The limit $\lim S_n$ exists if and only if $\lim a_{n+1}$ exists, giving the sum $\lim a_n - a_0$. □

Example 2: Telescoping Example

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. By partial fractions, $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. This is a telescoping series, so $\sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$.

Theorem 39: Riemann's Series and Integral Test

Let $f : [x_0, +\infty) \to (0, +\infty)$ be decreasing with $\lim_{x\to\infty} f(x) = 0$. Then:

$$\int_m^{n+1} f(x)dx \leq \sum_{k=m}^{n} f(k) \leq f(m) + \int_m^{n} f(x)dx$$

Using this on $f(x) = \frac{1}{x^s}$, we find that Riemann's Series $\sum_{n=1}^{\infty} \frac{1}{n^s}$ converges if and only if $s > 1$. The Riemann Zeta Function is denoted $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$.

PROOF. Because f is decreasing, for any $x \in [k, k+1]$, we have $f(k+1) \leq f(x) \leq f(k)$. Integrating this over the interval $[k, k+1]$ gives:

$$f(k+1) \leq \int_k^{k+1} f(x)dx \leq f(k)$$

Summing these inequalities from $k = m$ to $k = n$ gives:

$$\sum_{k=m}^{n} f(k+1) \leq \int_m^{n+1} f(x)dx \leq \sum_{k=m}^{n} f(k)$$

The right side gives $\int_m^{n+1} f(x)dx \leq \sum_{k=m}^{n} f(k)$. The left side gives $\sum_{k=m+1}^{n+1} f(k) \leq \int_m^{n+1} f(x)dx$, which implies $\sum_{k=m}^{n} f(k) \leq f(m) + \int_m^{n} f(x)dx$. For $f(x) = x^{-s}$, the integral $\int_1^n x^{-s}dx = \frac{n^{1-s}-1}{1-s}$ (for $s \neq 1$) converges as $n \to \infty$ if and only if $s > 1$. The sum converges iff the integral converges.
Explicit proof for divergence of Harmonic Series $\sum_{n=1}^{\infty} \frac{1}{n}$: Consider the difference of partial sums:

$$S_{2n} - S_n = \frac{1}{n+1} + \cdots + \frac{1}{2n}$$

There are n terms, and each term is $\geq \frac{1}{2n}$. Thus:

$$S_{2n} - S_n \geq n \cdot \frac{1}{2n} = \frac{1}{2}$$

Since the sequence of partial sums does not satisfy the Cauchy criterion (the difference between S_{2n} and S_n does not go to 0), the sequence of partial sums diverges. Thus, the harmonic series diverges. □

Theorem 40: Absolute vs. Alternating Series

The series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges. If a series converges absolutely, then it converges.

- **Alternating Series (Leibniz Test):** If (b_n) is a decreasing sequence of positive terms with $\lim b_n = 0$, then the series $\sum_{n=0}^{\infty}(-1)^n b_n$ converges.
- **Riemann's Rearrangement Theorem:** Let $a_n \in \mathbb{R}$. If $\sum a_n$ converges but $\sum |a_n|$ diverges (conditionally convergent), then for any $x \in \mathbb{R}$, there exists a permutation $\sigma : \mathbb{N} \to \mathbb{N}$ such that $\sum_{n=0}^{\infty} a_{\sigma(n)}$ converges to x.

Proof. (Absolute convergence implies convergence): Note that $0 \leq a_n + |a_n| \leq 2|a_n|$. Since $\sum |a_n|$ converges, $\sum 2|a_n|$ converges. By the Comparison Test, $\sum(a_n + |a_n|)$ converges. Since $a_n = (a_n + |a_n|) - |a_n|$, and both sums on the right converge, $\sum a_n$ must also converge. (Alternating Series): Let $S_n = \sum_{k=0}^{n}(-1)^k b_k$. $S_{2n+2} - S_{2n} = -b_{2n+1} + b_{2n+2} \leq 0$, so S_{2n} is decreasing. $S_{2n+3} - S_{2n+1} = b_{2n+2} - b_{2n+3} \geq 0$, so S_{2n+1} is increasing. $S_{2n} - S_{2n+1} = b_{2n+1} \to 0$. By the Adjacent Sequences Theorem, they converge to the same limit, so (S_n) converges. (Riemann's Rearrangement): Let $p_n = \max(a_n, 0)$ and $q_n = -\min(a_n, 0)$. Both $p_n \geq 0$ and $q_n \geq 0$, and $a_n = p_n - q_n$, $|a_n| = p_n + q_n$. Because $\sum a_n$ converges but $\sum |a_n|$ diverges, both $\sum p_n$ and $\sum q_n$ must diverge to $+\infty$. (If one converged, the other would have to converge for their sum/difference to converge). Because $\sum a_n$ converges, $a_n \to 0$, so $p_n \to 0$ and $q_n \to 0$. To rearrange to converge to x: take positive terms p_n sequentially just until the partial sum strictly exceeds x. (Possible because $\sum p_n = +\infty$). Then subtract negative terms q_n sequentially just until the partial sum falls strictly below x. Repeat this process indefinitely. Because $p_n \to 0$ and $q_n \to 0$, the "overshoot" amount tends to 0, meaning the sequence of rearranged partial sums converges exactly to x. □

4 Continuity

Definition 41: Closure and Function Limits

The closure of $A \subset \mathbb{R}$, denoted $\overline{A}$, is the set of all limits of sequences of elements of A. Equivalently (Neighborhood Characterization), $x \in \overline{A} \iff \forall \epsilon > 0, A \cap (x - \epsilon, x + \epsilon) \neq \emptyset$. Let $f : A \to \mathbb{R}$ and $x_0 \in \overline{A}$. $l \in \mathbb{R}$ is the limit of f as x tends to x_0 if:

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } \forall x \in A, \text{if } 0 < |x - x_0| < \delta \text{ then } |f(x) - l| < \epsilon$$

We write $l = \lim_{x \to x_0} f(x)$. The limit is unique if it exists. Equivalently (Sequential Criterion), f has a limit l as $x \to x_0$ iff for all sequences (a_n) in A converging to x_0, the sequence $(f(a_n))$ converges to l.

Definition 42: Infinite Limits for Functions

- $\lim_{x \to x_0} f(x) = +\infty \iff \forall M > 0, \exists \delta > 0$ such that $0 < |x - x_0| < \delta \implies f(x) > M$.

- $\lim_{x\to+\infty} f(x) = l \iff \forall \epsilon > 0, \exists M > 0$ such that $x > M \implies |f(x) - l| < \epsilon$.

Definition 43: Continuity at a Point

A function $f : A \to \mathbb{R}$ is continuous at $x_0 \in A$ if $\lim_{x\to x_0} f(x)$ exists and is equal to $f(x_0)$. This means $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall x \in A$, if $|x - x_0| < \delta$ then $|f(x) - f(x_0)| < \epsilon$.

Remark 44: Fixed Point Property

If a sequence is defined by a recurrence relation $x_{n+1} = f(x_n)$ and converges to a limit a, and f is continuous at a, then $a = f(a)$ (so a is a fixed point of f).

Remark 45: Properties of Continuous Functions

If f and g are continuous at x_0, then $f + g$, $f - g$, $f \cdot g$, and f/g (if $g(x_0) \neq 0$) are continuous at x_0. Compositions $(g \circ f)$, $|f|$, $\max(f, g)$, and $\min(f, g)$ are also continuous. A function is continuous on an interval if it is both left and right continuous everywhere on that interval. (Key Example: The Dirichlet function $I_{\mathbb{Q}}(x)$ which is 1 if $x \in \mathbb{Q}$ and 0 if $x \notin \mathbb{Q}$ is discontinuous everywhere.)

Theorem 46: Set of Periods

The set of periods of a continuous periodic function is either of the form $T\mathbb{Z}$ for some minimal period $T > 0$, or it is dense in $\mathbb{R}$ (e.g., constant functions).

Theorem 47: Global Properties of Continuous Functions

- **Intermediate Value Theorem (IVT):** If $f : [a, b] \to \mathbb{R}$ is continuous, then for every z strictly between $f(a)$ and $f(b)$, there exists $c \in (a, b)$ such that $f(c) = z$.
- **Extreme Value Theorem (EVT):** If $f : [a, b] \to \mathbb{R}$ is continuous, then $f([a, b]) = [m, M]$ where $m = \min f$ and $M = \max f$. That is, f is bounded and attains its bounds.

Proof. (Intermediate Value Theorem): Assume without loss of generality that $f(a) < z < f(b)$. Let $S = \{x \in [a, b] : f(x) \leq z\}$. The set S is non-empty (since $a \in S$) and bounded above by b. By the least upper bound property, $c = \sup S$ exists, and $c \in [a, b]$. We claim that $f(c) = z$. We prove this using the sequential criterion for continuity. First, by definition of the supremum, there exists a sequence (x_n) in S such that $x_n \to c$. Since $x_n \in S$, we know $f(x_n) \leq z$ for all n. By continuity of f, $\lim_{n\to\infty} f(x_n) = f(c)$. Thus, $f(c) \leq z$. Second, consider a sequence (y_n) defined by $y_n = c + \frac{1}{n}$. For sufficiently large n, $y_n \in (c, b]$. Since $y_n > c = \sup S$, $y_n \notin S$, which means $f(y_n) > z$. As $n \to \infty$, $y_n \to c$. By continuity of f, $\lim_{n\to\infty} f(y_n) = f(c)$. Thus, $f(c) \geq z$. Since $f(c) \leq z$ and $f(c) \geq z$, it must be that $f(c) = z$.
(Extreme Value Theorem): Step 1 (Boundedness): Suppose f is not bounded on $[a, b]$. Then for each $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ such that $|f(x_n)| > n$. Since (x_n) is a bounded sequence in $[a, b]$, the Bolzano-Weierstrass theorem guarantees a convergent subsequence (x_{n_k}) that converges to some limit $c \in [a, b]$. Since f is continuous at c, $\lim_{k\to\infty} f(x_{n_k}) = f(c)$. However, $|f(x_{n_k})| > n_k \to \infty$, which contradicts that the sequence of function values converges to a

finite real number $f(c)$. Thus, f is bounded.
Step 2 (Attaining the bounds): Since $f([a,b])$ is bounded, it has a supremum, say $M = \sup f([a,b])$. By the definition of supremum, for any $n \in \mathbb{N}$, $M - \frac{1}{n}$ is not an upper bound, so there exists $x_n \in [a,b]$ such that $M - \frac{1}{n} < f(x_n) \leq M$. By the Squeeze Theorem, $\lim_{n\to\infty} f(x_n) = M$. By Bolzano-Weierstrass, the sequence (x_n) has a convergent subsequence (x_{n_k}) converging to some $c \in [a,b]$. Since f is continuous at c, $f(c) = \lim_{k\to\infty} f(x_{n_k}) = M$. Therefore, f attains its maximum at c. The proof for the minimum m follows identically by taking the infimum. □

Definition 48: Uniform Continuity

A function $f : I \to \mathbb{R}$ is uniformly continuous on the interval I if:

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } \forall x, y \in I, \text{ if } |x - y| < \delta \text{ then } |f(x) - f(y)| < \epsilon$$

(Note: Unlike standard continuity where δ depends on both ϵ and the specific point x_0, in uniform continuity, a single δ works for all points x, y simultaneously).

5 Differentiability

Definition 49: Derivative and C^n Functions

Let $I = (a,b)$ be an open interval, $f : (a,b) \to \mathbb{R}$, and $x_0 \in (a,b)$. f is differentiable at x_0 if the quotient $\frac{f(x)-f(x_0)}{x-x_0}$ has a limit as $x \to x_0$. This limit is the derivative, denoted $f'(x_0)$. Equivalently, there exists a function $\epsilon(h)$ with $\lim_{h\to 0} \epsilon(h) = 0$ such that $f(x_0 + h) = f(x_0) + f'(x_0)h + h\epsilon(h)$. C^n **Functions:** A function is of class C^n on an interval if it is n-times differentiable and its n-th derivative is continuous.

Proposition 50: Differentiability Properties

- If f is differentiable at x_0, then f is continuous at x_0.
- f is differentiable at x_0 iff it is left and right differentiable at x_0 and $f'_+(x_0) = f'_-(x_0)$.
- $(f+g)' = f' + g'$, $(fg)' = f'g + fg'$, and $(1/g)' = -g'/g^2$ (where $g \neq 0$).
- **Chain Rule:** If f is differentiable at x_0 and g is differentiable at $f(x_0)$, then $g \circ f$ is differentiable at x_0 and $(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0)$.

Proof. (Diff $\implies$ Cont): $\lim_{x\to x_0}(f(x) - f(x_0)) = \lim_{x\to x_0} \frac{f(x)-f(x_0)}{x-x_0}(x - x_0) = f'(x_0) \cdot 0 = 0$.
(Product Rule): $\lim_{x\to x_0} \frac{f(x)g(x)-f(x_0)g(x_0)}{x-x_0} = \lim_{x\to x_0} \left[g(x)\frac{f(x)-f(x_0)}{x-x_0} + f(x_0)\frac{g(x)-g(x_0)}{x-x_0}\right] = g(x_0)f'(x_0) + f(x_0)g'(x_0)$.
(Quotient Rule for $1/g$):

$$\lim_{x\to x_0} \frac{\frac{1}{g(x)} - \frac{1}{g(x_0)}}{x - x_0} = \lim_{x\to x_0} \frac{g(x_0) - g(x)}{g(x)g(x_0)(x - x_0)} = \lim_{x\to x_0} \frac{-1}{g(x)g(x_0)} \cdot \frac{g(x) - g(x_0)}{x - x_0} = \frac{-g'(x_0)}{g(x_0)^2}$$

(Chain Rule): Define a function $h(y) = \frac{g(y)-g(y_0)}{y-y_0}$ for $y \neq y_0$, and $h(y_0) = g'(y_0)$. Since g is differentiable at y_0, h is continuous at y_0. For all y, $g(y) - g(y_0) = h(y)(y - y_0)$. Substitute $y = f(x)$ and $y_0 = f(x_0)$:

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = h(f(x))\frac{f(x) - f(x_0)}{x - x_0}$$

Taking the limit as $x \to x_0$. Because f is continuous, $f(x) \to f(x_0)$, so $h(f(x)) \to h(f(x_0)) = g'(f(x_0))$. The result is $g'(f(x_0))f'(x_0)$. □

Theorem 51: Inverse Function Theorem

Let I be an open interval and $f : I \to J$ (where $J = f(I)$) be continuous and strictly monotone. Then f is bijective and $f^{-1} : J \to I$ is strictly monotone and continuous. Furthermore, if f is differentiable at $x_0 \in I$ and $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$.

Proof. (Bijectivity and Continuity): Since f is strictly monotone, it is injective. By the Intermediate Value Theorem, since f is continuous on the interval I, its image $J = f(I)$ is also an interval. Thus, $f : I \to J$ is surjective, making it a bijection. Let $f^{-1} : J \to I$. Because f is strictly monotone, f^{-1} preserves this monotonicity. A monotone function that maps an interval onto an interval must be continuous; if it had a jump discontinuity, its image would miss values, contradicting that its image is the interval I. Thus f^{-1} is continuous.
(Derivative): Let $y \in J$ with $y \neq y_0 = f(x_0)$. Since f is bijective, $\exists! x \in I$ such that $f(x) = y$. As $y \to y_0$, by continuity of f^{-1}, $x \to x_0$.

$$\lim_{y \to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{x \to x_0} \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}} = \frac{1}{f'(x_0)}$$

□

Theorem 52: Extrema and Mean Value Theorems

An extremum is a point where f achieves a local max or min. If f has an extremum at $x_0 \in I$ (an open interval) and is differentiable there, then $f'(x_0) = 0$ (Fermat's Theorem).

- **Rolle's Theorem:** If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.
- **Mean Value Theorem (MVT):** If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a)$.
- **Lipschitz Corollary:** $|f(b) - f(a)| \leq \left(\sup_{(a,b)} |f'|\right) |b - a|$.

A differentiable function f on an open interval I is increasing if and only if $f'(x) \geq 0$ for all $x \in I$.

Proof. (Fermat's Theorem): Suppose f has a local maximum at x_0. For small $h > 0$, $f(x_0 + h) \leq f(x_0)$, so $\frac{f(x_0+h)-f(x_0)}{h} \leq 0$. Taking the limit as $h \to 0^+$ gives $f'(x_0) \leq 0$. For small $h < 0$, $f(x_0 + h) \leq f(x_0)$, but h is negative, so $\frac{f(x_0+h)-f(x_0)}{h} \geq 0$. Taking the limit as $h \to 0^-$ gives $f'(x_0) \geq 0$. Since f is differentiable at x_0, the left and right limits must be equal. Thus, $f'(x_0) = 0$.
(Rolle's Theorem): By the Extreme Value Theorem, f attains a global maximum M and minimum m on $[a, b]$. If $M = m$, f is constant, so $f'(x) = 0$ for all $x \in (a, b)$. If $M > m$, since $f(a) = f(b)$, f must attain an extremum at some point c strictly inside (a, b). By Fermat's Theorem, $f'(c) = 0$.
(Mean Value Theorem): Define $g(x) = f(x) - \left(f(a) + \frac{f(b)-f(a)}{b-a}(x - a)\right)$. g satisfies the hypotheses of Rolle's Theorem because $g(a) = g(b) = 0$. Thus, there exists $c \in (a, b)$ where $g'(c) = 0$. Since $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$, setting $g'(c) = 0$ yields $f'(c) = \frac{f(b)-f(a)}{b-a}$.

(Monotonicity via MVT): If $f'(x) \geq 0$ for all $x \in I$, pick any $x_1 < x_2$ in I. By MVT, there is $c \in (x_1, x_2)$ such that $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. Since $f'(c) \geq 0$ and $x_2 - x_1 > 0$, we have $f(x_2) - f(x_1) \geq 0$, so f is increasing. □

Example 3: Sine Lipschitz Example

For $f(x) = \sin x$, $f'(x) = \cos x$. Since $|\cos x| \leq 1$, we can apply the Lipschitz corollary of the MVT to get $|\sin y - \sin x| \leq |y - x|$ for all $x, y \in \mathbb{R}$.

6 Transcendental and Convex Functions

Remark 53: Trigonometric Functions

Sine and cosine are n-times differentiable for all $n \in \mathbb{N}$. Complex exponential expressions: $\cos x = \frac{e^{ix}+e^{-ix}}{2}$ and $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$.

- $\cos(x + y) = \cos x \cos y - \sin x \sin y$ and $\sin(x + y) = \sin x \cos y + \sin y \cos x$.
- $\arcsin : (-1, 1) \to (-\frac{\pi}{2}, \frac{\pi}{2})$ has derivative $\frac{1}{\sqrt{1-x^2}}$.
- $\arccos : (-1, 1) \to (0, \pi)$ has derivative $\frac{-1}{\sqrt{1-x^2}}$.
- $\tan x = \frac{\sin x}{\cos x}$ has derivative $\frac{1}{\cos^2 x} = 1 + \tan^2 x$. $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$ has derivative $\frac{1}{1+x^2}$.

Proof. (Derivative of arcsin): Let $y = \arcsin x$ for $x \in (-1, 1)$. Then $x = \sin y$ where $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$. By the Inverse Function Theorem: $\arcsin'(x) = \frac{1}{\sin' y} = \frac{1}{\cos y}$. Since $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\cos y > 0$, so $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$. Thus, $\arcsin'(x) = \frac{1}{\sqrt{1-x^2}}$.
(Derivative of arccos): Similarly, let $y = \arccos x \in (0, \pi)$. Then $x = \cos y$. $\arccos'(x) = \frac{1}{\cos' y} = \frac{-1}{\sin y}$. Since $\sin y > 0$ on $(0, \pi)$, $\sin y = \sqrt{1 - \cos^2 y} = \sqrt{1 - x^2}$. Thus $\arccos'(x) = \frac{-1}{\sqrt{1-x^2}}$.
(Derivative of tan): By the quotient rule: $\tan' x = \left(\frac{\sin x}{\cos x}\right)' = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = 1 + \tan^2 x$.
(Derivative of arctan): Let $y = \arctan x$. Then $x = \tan y$. By the Inverse Function Theorem: $\arctan'(x) = \frac{1}{\tan' y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$. □

Definition 54: Natural Log and Exponential

For $x > 0$, the natural logarithm is defined via the integral:

$$\ln x = \int_1^x \frac{1}{t} dt$$

The inverse function is the exponential function $\exp : \mathbb{R} \to (0, +\infty)$. Euler's number is defined as $e = \exp(1)$. General power functions are defined as $x^a = \exp(a \ln x)$ for $x > 0$, with derivative $(x^a)' = ax^{a-1}$. General exponential functions are $a^x = \exp(x \ln a)$, with derivative $(a^x)' = (\ln a)a^x$.

Theorem 55: Properties of Log and Exp

- $\ln : (0, +\infty) \to \mathbb{R}$ is strictly increasing, and $\ln(xy) = \ln x + \ln y$.
- $\exp'(x) = \exp(x)$ and $\exp(x + y) = \exp(x)\exp(y)$.
- $\lim_{x\to+\infty} \ln x = +\infty$.

PROOF. (Properties of ln): By the Fundamental Theorem of Calculus, $\ln'(x) = \frac{1}{x}$. Since $\frac{1}{x} > 0$ for $x > 0$, the function $\ln x$ is strictly increasing. For the product rule, let $f(x) = \ln(ax)$ for a fixed $a > 0$. By the chain rule, $f'(x) = \frac{1}{ax} \cdot a = \frac{1}{x} = \ln'(x)$. Since their derivatives are equal on an interval, they differ by a constant: $f(x) = \ln x + C$. To find C, evaluate at $x = 1$: $f(1) = \ln(a) = \ln(1) + C = 0 + C$, so $C = \ln a$. Thus, $\ln(ax) = \ln x + \ln a$.
(Derivative of exp): Let $y = \exp(x)$, which means $x = \ln y$. By the Inverse Function Theorem:

$$\exp'(x) = \frac{1}{\ln'(y)} = \frac{1}{\frac{1}{y}} = y = \exp(x)$$

(Limit of ln at $+\infty$): We use a harmonic sum lower bound. For an integer $n > 1$:

$$\ln n = \int_1^n \frac{1}{t}dt = \sum_{k=1}^{n-1} \int_k^{k+1} \frac{1}{t}dt \geq \sum_{k=1}^{n-1} \frac{1}{k+1} = \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

As this is the tail of the harmonic series which diverges to $+\infty$, the sequence $\ln n \to +\infty$. Since ln is strictly increasing, $\lim_{x\to+\infty} \ln x = +\infty$. □

Lemma 56: Hierarchy of Infinities (Growth Comparison)

For any $a > 0$, $\lim_{x\to+\infty} \frac{e^x}{x^a} = +\infty$ and $\lim_{x\to+\infty} \frac{\ln x}{x^a} = 0$.

Remark 57: Hyperbolic Functions

The hyperbolic sine and cosine are defined as $\cosh x = \frac{e^x+e^{-x}}{2}$ and $\sinh x = \frac{e^x-e^{-x}}{2}$. They satisfy the fundamental identity $\cosh^2 x - \sinh^2 x = 1$. **Addition formulas:** $\cosh(a + b) = \cosh a \cosh b + \sinh a \sinh b$, and $\sinh(a + b) = \sinh a \cosh b + \cosh a \sinh b$. **Derivative of tanh:** $\tanh' x = \frac{1}{\cosh^2 x} = 1 - \tanh^2 x$. Their inverses include $\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$ (for $x \geq 1$) and $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$.

Theorem 58: Convex Functions

A function $f : (a, b) \to \mathbb{R}$ is convex if for all $x, y \in (a, b)$ and $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

If f is differentiable on (a, b), f is convex if and only if f' is increasing on (a, b). If f is twice differentiable, f is convex if and only if $f''(x) \geq 0$ for all $x \in (a, b)$.

PROOF. (Twice differentiable case): Suppose $f'' \geq 0$. This implies f' is an increasing function. We want to show f is convex. Let $x < y$ in (a, b) and let $t \in (0, 1)$. Define $z = tx + (1 - t)y$. Notice that $x < z < y$. By the Mean Value Theorem applied to the intervals $[x, z]$ and $[z, y]$,

there exist points $c_1 \in (x, z)$ and $c_2 \in (z, y)$ such that:

$$\frac{f(z) - f(x)}{z - x} = f'(c_1) \quad \text{and} \quad \frac{f(y) - f(z)}{y - z} = f'(c_2)$$

Since $c_1 < z < c_2$ and f' is increasing, we have $f'(c_1) \leq f'(c_2)$. Therefore:

$$\frac{f(z) - f(x)}{z - x} \leq \frac{f(y) - f(z)}{y - z}$$

Substitute $z - x = (1 - t)(y - x)$ and $y - z = t(y - x)$ into the denominators:

$$\frac{f(z) - f(x)}{(1 - t)(y - x)} \leq \frac{f(y) - f(z)}{t(y - x)}$$

Since $t \in (0, 1)$ and $y > x$, we can multiply both sides by $t(1 - t)(y - x) > 0$ to get:

$$t(f(z) - f(x)) \leq (1 - t)(f(y) - f(z))$$

Expanding and rearranging terms:

$$tf(z) - tf(x) \leq f(y) - f(z) - tf(y) + tf(z)$$

$$f(z) \leq tf(x) + (1 - t)f(y)$$

Substituting back $z = tx + (1 - t)y$, we recover exactly the definition of convexity:

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

The reverse implication (convex $\implies f'' \geq 0$) follows by reversing these steps using limits. □

7 Taylor Expansions

Definition 59: Taylor Expansion

Let $f : (a, b) \to \mathbb{R}$ and $x_0 \in (a, b)$. f admits at x_0 a Taylor expansion (or Taylor polynomial with Peano remainder) of order n if there exists a polynomial $P \in \mathbb{R}[X]$ with $\deg P \leq n$ and a function $\Theta : (a, b) \to \mathbb{R}$ such that:

$$f(x) = P(x - x_0) + (x - x_0)^n \Theta(x) \quad \text{and} \quad \lim_{x \to x_0} \Theta(x) = 0$$

If a Taylor expansion exists, it is unique, and its coefficients are given by $c_k = \frac{f^{(k)}(x_0)}{k!}$.
Note on Orders: An order 0 Taylor expansion at x_0 is equivalent to continuity at x_0. An order 1 Taylor expansion is equivalent to differentiability at x_0.

Proof. (Uniqueness of the Taylor Expansion): Suppose f has two Taylor expansions of order n at x_0: $f(x) = P_1(x - x_0) + (x - x_0)^n \Theta_1(x)$ and $f(x) = P_2(x - x_0) + (x - x_0)^n \Theta_2(x)$, where $\lim_{x \to x_0} \Theta_1(x) = \lim_{x \to x_0} \Theta_2(x) = 0$. Let $Q(h) = P_1(h) - P_2(h) = c_0 + c_1 h + \cdots + c_n h^n$. Subtracting the two expansions gives:

$$0 = Q(x - x_0) + (x - x_0)^n (\Theta_1(x) - \Theta_2(x))$$

Taking the limit as $x \to x_0$, we get $Q(0) = 0$, so $c_0 = 0$. Dividing by $x - x_0$ and taking the limit gives $c_1 = 0$, and iterating this process yields $c_k = 0$ for all $0 \leq k \leq n$. Thus $P_1 = P_2$, proving uniqueness. □

Remark 60: Cautionary Counterexample

Having a Taylor expansion of order n at x_0 does not imply that f is n-times differentiable at x_0. For example, $f(x) = |x|^{\alpha+2}\cos(1/x)$ (with $f(0) = 0$) for $\alpha > 0$ admits a Taylor expansion at 0, but may not be twice differentiable at 0.

Theorem 61: Taylor-Lagrange Theorem

Assume f is n-times continuously differentiable on (a, b), and $n+1$ times differentiable on (a, b). Fix $x_0 \in (a, b)$. For all $x \in (a, b)$ with $x \neq x_0$, there exists c strictly between x_0 and x such that:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$$

Second Derivative Test: If $f'(x_0) = 0$ and $f''(x_0) > 0$, then x_0 is a point where f achieves a local minimum.

Proof. (Taylor-Lagrange Formula): Fix $x \in (a, b)$. Let M be the unique real number such that:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + M(x - x_0)^{n+1}$$

We want to show that $M = \frac{f^{(n+1)}(c)}{(n+1)!}$ for some c between x_0 and x. Define an auxiliary function $g(t)$ for t between x_0 and x:

$$g(t) = f(x) - \left(\sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k + M(x - t)^{n+1} \right)$$

Notice that $g(x) = f(x) - f(x) - 0 = 0$, and $g(x_0) = 0$ by our choice of M. Since $g(x) = g(x_0) = 0$, by Rolle's Theorem, there exists c strictly between x_0 and x such that $g'(c) = 0$. Let's compute $g'(t)$. The sum telescopes when differentiated:

$$\frac{d}{dt}\left(\sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k \right) = \sum_{k=0}^{n} \left(\frac{f^{(k+1)}(t)}{k!}(x - t)^k - \frac{f^{(k)}(t)}{(k-1)!}(x - t)^{k-1} \right) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n$$

So, differentiating $g(t)$ yields:

$$g'(t) = -\frac{f^{(n+1)}(t)}{n!}(x - t)^n + (n+1)M(x - t)^n$$

Setting $g'(c) = 0$ gives:

$$\frac{f^{(n+1)}(c)}{n!}(x - c)^n = (n+1)M(x - c)^n$$

Since $c \neq x$, we can divide by $(n+1)(x - c)^n$ to get $M = \frac{f^{(n+1)}(c)}{(n+1)!}$.
(Second Derivative Test): Using the Taylor expansion of order 2 with the Peano remainder around x_0:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + (x - x_0)^2\Theta(x)$$

Since $f'(x_0) = 0$, this simplifies to:

$$f(x) - f(x_0) = (x - x_0)^2 \left(\frac{f''(x_0)}{2} + \Theta(x) \right)$$

Since $\lim_{x\to x_0}\Theta(x)=0$ and $f''(x_0)>0$, for x sufficiently close to x_0, the term in the parentheses is strictly positive. Because $(x-x_0)^2\geq 0$, it follows that $f(x)-f(x_0)\geq 0 \implies f(x)\geq f(x_0)$ in a neighborhood of x_0. Thus, f has a local minimum at x_0. □

Example 4: Applications of Taylor-Lagrange

Cosine Bound: Applying Taylor-Lagrange to $\cos x$ at $x_0=0$ to order 1: $\cos x = 1-\frac{\cos c}{2}x^2$ for some c between 0 and x. Since $|\cos c|\leq 1$, we get $|\cos x-1|\leq \frac{x^2}{2}$. **Limit Evaluation:** By the bound above, $\frac{-x^2}{2}\leq \cos x-1\leq \frac{x^2}{2}$. For $x\neq 0$, dividing by x^2 and taking limits gives $\lim_{x\to 0}\frac{\cos x-1}{x^2}=-\frac{1}{2}$.

Example 5: Common Taylor Series

- **Geometric:** $\frac{1}{1-x}=\sum_{k=0}^{\infty}x^k$ for $|x|<1$.
- **Alternating Geometric:** $\frac{1}{1+x}=\sum_{k=0}^{\infty}(-1)^k x^k$ for $|x|<1$.
- **Squared Geometric:** $\frac{1}{1-x^2}=\sum_{k=0}^{\infty}x^{2k}$ for $|x|<1$.
- **Exponential:** $\exp(x)=\sum_{k=0}^{+\infty}\frac{x^k}{k!}$, implying $e=\sum_{k=0}^{\infty}\frac{1}{k!}$.
- **Sine:** $\sin x=\frac{x}{1!}-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\dots$

Proof. (Geometric Series): The Taylor expansion of $f(x)=(1-x)^{-1}$ at $x_0=0$ yields $f^{(k)}(0)=k!$. Thus the Taylor series is $\sum x^k$. We already know from the formula for finite geometric sums that $S_n=\frac{1-x^{n+1}}{1-x}$. For $|x|<1$, $\lim_{n\to\infty}x^{n+1}=0$, so the series converges exactly to $\frac{1}{1-x}$.
(Exponential Series): $f^{(k)}(0)=1$ for all k. By Taylor-Lagrange: $\exp(x)=\sum_{k=0}^{n}\frac{x^k}{k!}+\frac{\exp(c)}{(n+1)!}x^{n+1}$. For fixed x, as $n\to\infty$, the remainder $\left|\frac{\exp(c)x^{n+1}}{(n+1)!}\right|\leq\frac{\max(1,e^x)|x|^{n+1}}{(n+1)!}\to 0$. So it converges to $\exp(x)$ everywhere.
(Sine Series): Let $f(x)=\sin x$. The derivatives cycle: $\cos x,-\sin x,-\cos x,\sin x$. At $x_0=0$, the values are $1,0,-1,0$. This gives the odd-power series. By Taylor-Lagrange, the remainder is $\frac{f^{(n+1)}(c)}{(n+1)!}x^{n+1}$. Since all derivatives of sine are bounded by 1 in absolute value ($|\sin c|\leq 1,|\cos c|\leq 1$), the remainder is bounded by $\frac{|x|^{n+1}}{(n+1)!}$. For any fixed $x\in\mathbb{R}$, $\frac{|x|^{n+1}}{(n+1)!}\to 0$ as $n\to\infty$. Thus the Taylor series converges to $\sin x$ for all x. □